WORKED SOLUTION

STEP III 2015, Question 4

Pure • Roots of Cubic Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q4

The key idea

The three conditions in part (iii) look like a trigonometric puzzle, but each is the imaginary part of something: $r_k^n \sin(n\theta_k)$ is the imaginary part of z_k^n when $z_k = r_k e^{i\theta_k}$. So the conditions say that the power sums S_1, S_2, S_3 of three complex numbers are real. Part (ii) turns real power sums into a cubic with real coefficients, and part (i) then hands us a real root.

Solution

Part (i)

Let $f(x) = x^3 + ax^2 + bx + c$ for real x; this is a continuous real function. Put $R = 1 + |a| + |b| + |c|$. For $x \geq R$ we have $x \geq 1$, so $|x| \leq x^2$ and $1 \leq x^2$, giving $|ax^2 + bx + c| \leq (|a| + |b| + |c|)x^2$. Hence

$$f(x) \geq x^3 - (|a| + |b| + |c|)x^2 = x^2(x - |a| - |b| - |c|) > 0.$$

In the same way, for $x \leq -R$ we get $f(x) \leq -|x|^2(|x| - |a| - |b| - |c|) < 0$. So $f(-R) < 0 < f(R)$, and by the intermediate value theorem f has a zero between $-R$ and R. That zero is a real root of $(*)$.

(Equivalently: non-real roots of a real polynomial come in conjugate pairs, and three roots cannot all be paired off.)

Part (ii)

Since z_1, z_2, z_3 are the roots,

$$z^3 + az^2 + bz + c = (z - z_1)(z - z_2)(z - z_3),$$

and comparing coefficients gives

$$a = -(z_1 + z_2 + z_3), \qquad b = z_1z_2 + z_2z_3 + z_3z_1, \qquad c = -z_1z_2z_3.$$

So $a = -S_1$. Squaring the sum, $S_1^2 = S_2 + 2(z_1z_2 + z_2z_3 + z_3z_1) = S_2 + 2b$, so

$$a = -S_1, \qquad b = \tfrac{1}{2}(S_1^2 - S_2).$$

For c, rather than expanding S_1^3, use the fact that each root satisfies the equation: $z_k^3 + az_k^2 + bz_k + c = 0$ for $k = 1, 2, 3$. Adding these three equations,

$$S_3 + aS_2 + bS_1 + 3c = 0.$$

Substituting for a and b and multiplying by 2:

$$\begin{aligned} 6c &= -2S_3 + 2S_1S_2 - S_1(S_1^2 - S_2) \\ &= -S_1^3 + 3S_1S_2 - 2S_3, \end{aligned}$$

as required.

Part (iii)

Define $z_k = r_k e^{i\theta_k}$ for $k = 1, 2, 3$. By de Moivre's theorem $z_k^n = r_k^n(\cos n\theta_k + i \sin n\theta_k)$, so the three given conditions say exactly

$$\operatorname{Im} S_1 = \operatorname{Im} S_2 = \operatorname{Im} S_3 = 0,$$

where S_1, S_2, S_3 are the power sums of z_1, z_2, z_3 as in part (ii). So S_1, S_2, S_3 are all real.

Ask yourself Part (ii) started from a cubic. Where is the cubic here?

We build it: let a, b, c be defined by $(z - z_1)(z - z_2)(z - z_3) = z^3 + az^2 + bz + c$. Then z_1, z_2, z_3 are the roots of this cubic, so part (ii) applies and gives

$$a = -S_1, \qquad b = \tfrac{1}{2}(S_1^2 - S_2), \qquad c = \tfrac{1}{6}(-S_1^3 + 3S_1S_2 - 2S_3),$$

all of which are real. By part (i) the cubic has a real root, and its only roots are z_1, z_2, z_3, so some z_k is real. Now $z_k = r_k e^{i\theta_k}$ with $r_k > 0$ is real only if $\sin\theta_k = 0$, that is θ_k is a multiple of π; the only such value in $-\pi < \theta_k < \pi$ is $\theta_k = 0$. So $\theta_k = 0$ for at least one k.

Now suppose $\theta_1 = 0$, so $z_1 = r_1$ is real and non-zero. Then

$$z_2 + z_3 = S_1 - z_1 \quad \text{and} \quad z_2z_3 = \frac{-c}{z_1}$$

are both real. Hence z_2 and z_3 are the roots of the quadratic $z^2 - (z_2 + z_3)z + z_2z_3 = 0$, which has real coefficients. There are two cases.

- If the roots are non-real, they are complex conjugates: $z_3 = \overline{z_2} = r_2 e^{-i\theta_2}$. Since $-\theta_2$ also lies in $(-\pi, \pi)$ and the argument in that interval is unique, $\theta_3 = -\theta_2$ (and $r_3 = r_2$).
- If the roots are real, then exactly as above (each has positive modulus and argument in $(-\pi, \pi)$) we get $\theta_2 = \theta_3 = 0$, so again $\theta_2 = -\theta_3$.

In either case $\theta_2 = -\theta_3$, as required.

Checks

- Part (ii) with roots $1, 2, 3$: $S_1 = 6$, $S_2 = 14$, $S_3 = 36$, and $-216 + 252 - 72 = -36 = 6c$ with $c = -6$, as it should be for $(z-1)(z-2)(z-3)$.
- Part (iii) the other way round: with $\theta_1 = 0$, $\theta_3 = -\theta_2$ and $r_3 = r_2$, the second and third terms of each sum cancel and the first is zero, so all three conditions hold. A computer search solving the three conditions numerically from thousands of random starting points found only solutions with one angle zero and the other two opposite.

Common slips

- In part (i), saying "a cubic always crosses the axis" without justification; the sign change at large $|x|$ needs a line of argument.
- In part (iii), forgetting that $\theta_k = \pm\pi$ would also make z_k real; it is the open interval $(-\pi, \pi)$ that rules this out.
- Concluding $\theta_2 = -\theta_3$ only from the conjugate case and overlooking the case where z_2, z_3 are both real.

Beyond STEP

The coefficients of a polynomial are the elementary symmetric functions of its roots, and the identities in part (ii) are the first of Newton's identities linking them to the power sums. A first course in algebra shows that every symmetric polynomial in the roots is a polynomial in the elementary symmetric functions, which is the starting point of Galois theory. Allenby's *Rings, Fields and Groups* is a good place to read about this.

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